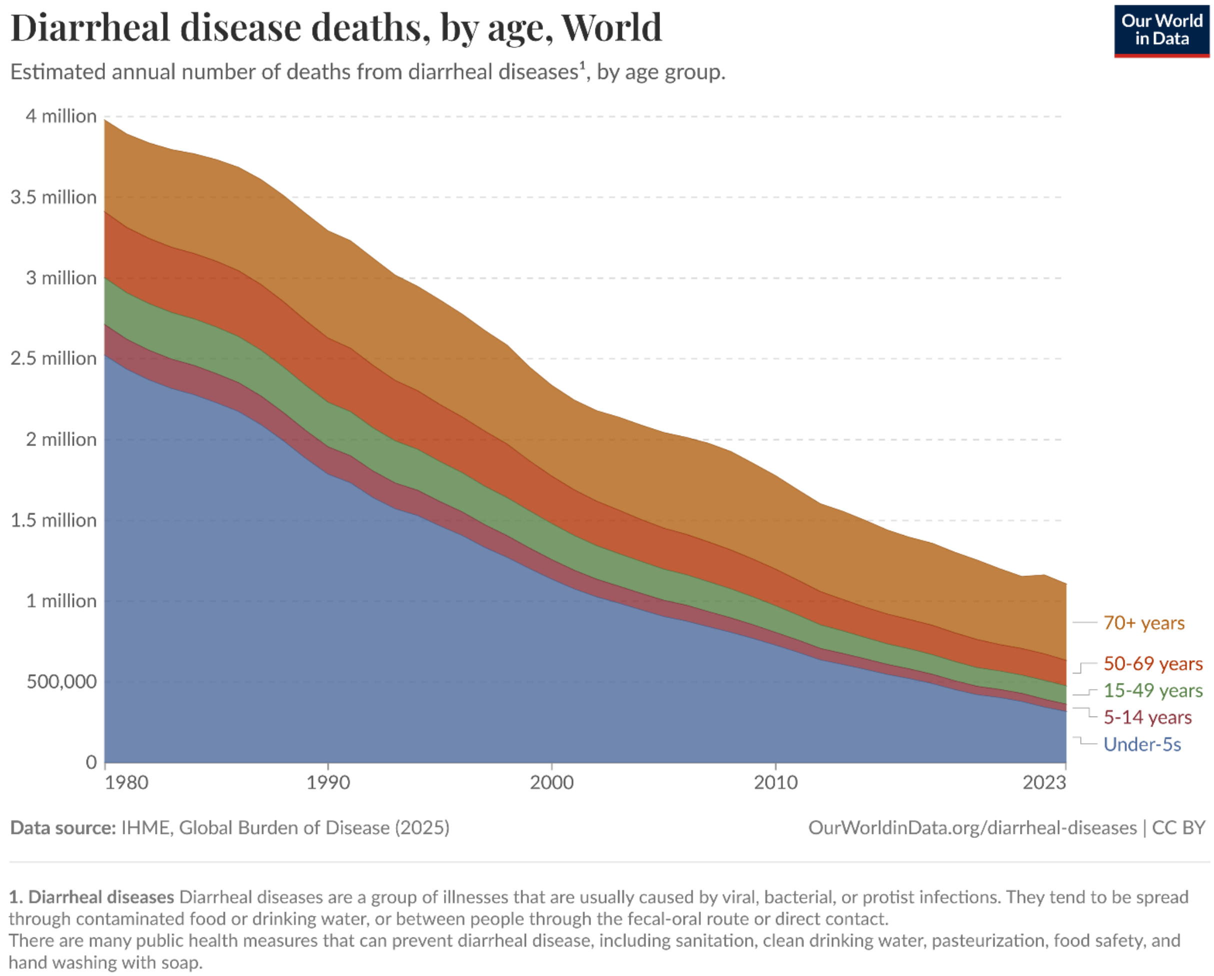
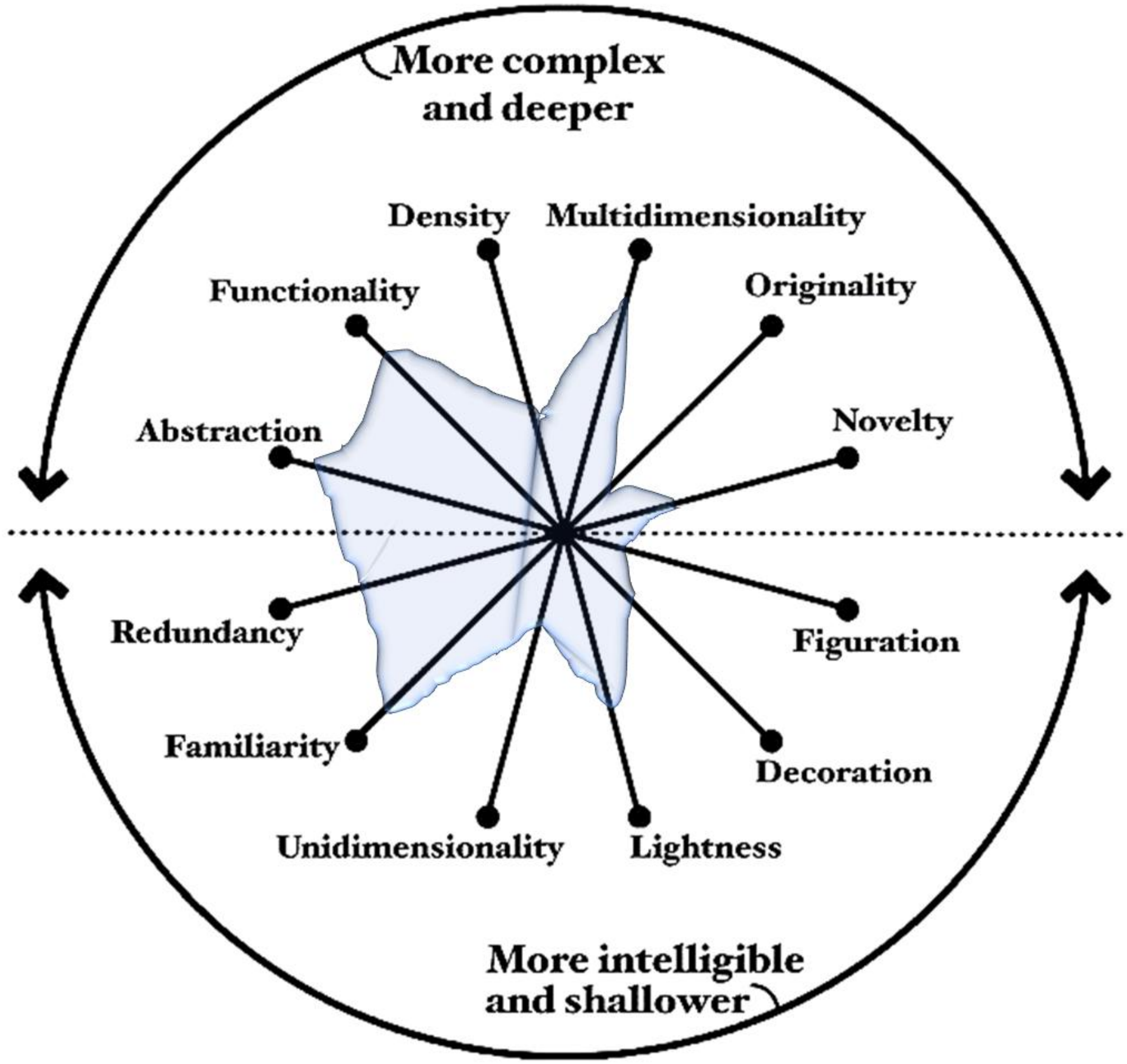


1. Main visualization goal and audience

The chart targets a broad, non-specialist public audience — the kind reached by Our World in Data's mission of making global development data accessible. The goal is explicitly communicative: to show, at a glance, that global deaths from diarrheal diseases have fallen dramatically over four decades, and that children under 5 were historically the dominant victims but that reduction has been the greatest success story. The stacked area form prioritizes emotional impact (the collapsing blue mass) over precise readout of individual values.

2. Visualization Dimensions Using Alberto Cairo's Visualization Wheel

The visualization *"Diarrheal Disease Deaths, by Age, World"* can be analyzed using Alberto Cairo's Visualization Wheel, which evaluates the balance between opposing design dimensions



Dimension Pair	Assessment
Abstraction vs. Figuration	**95% Abstraction – 5% Figuration**
Functionality vs. Decoration	**90% Functionality – 10% Decoration**
Density vs. Lightness	**40% Density – 60% Lightness**
Multidimensionality vs. Unidimensionality	**70% Multidimensionality – 30% Unidimensionality**
Originality vs. Familiarity	**20% Originality – 80% Familiarity**
Novelty vs. Redundancy	**30% Novelty – 70% Redundancy**

The chart strongly favors abstraction because it uses a statistical representation rather than pictorial elements. It is highly functional, as nearly all graphical elements contribute to communicating the data. The design leans toward lightness, presenting a large amount of information without overwhelming the reader. It is moderately multidimensional, displaying mortality counts, age groups, and temporal changes simultaneously. The visualization is highly familiar, employing a standard stacked area chart that most readers can easily understand. Finally, it contains a degree of redundancy through labels, colors, and ordering that reinforce comprehension.

3. Main Drawbacks Following Alberto Cairo and Edward Tufte's Principles

Although the visualization is effective, several limitations can be identified based on the recommendations of Alberto Cairo and Edward Tufte.

- **Difficult disaggregation:** Stacked areas make it nearly impossible to read the *absolute* values of middle layers (50–69 years, 15–49 years) accurately, since their baseline shifts with the layer below. The reader can see trends but not *values*. Cairo would flag this as a trade-off between the "whole picture" and individual accuracy.
- **Color accessibility:** The palette — a brownish orange, a warm orange-red, a muted green, a muted pink, and a steel blue — does not follow colorblind-safe conventions. The 5–14 and 15–49 bands (pink and green) are particularly hard to distinguish for deuteranopic viewers, violating Cairo's principle of accessibility.
- **No reference lines or annotations:** The chart misses opportunities to mark key events that would transform a trend chart into an *insight* chart, which Cairo would consider a missed step toward enlightenment.
- **Limited Exploration:** The visualization is primarily explanatory rather than exploratory. Users can identify the main message—the decline in diarrheal disease deaths—but have limited opportunities to investigate detailed patterns or relationships. trends

Edward Tufte would focus on data-ink ratio and graphical integrity:

- **The stacked form inflates visual complexity:** Tufte generally prefers small multiples over stacking, because each area's shape is distorted by the layers beneath it. A panel-per-age-group would preserve each individual trend more honestly.

- **Lie factor / perceptual distortion:** The area of the under-5 band visually dominates (it is the bottom, widest band with a stable baseline), which correctly reflects the data. But the upper bands appear to show almost no change because their *relative* share of a shrinking total remains roughly constant. This can mislead a reader into thinking, say, deaths among 50–69-year-olds barely changed, when in absolute terms they also fell substantially before 2010.

- **Good data-ink ratio overall:** Tufte would approve of the absence of gridline clutter, the use of dashed horizontal guides at meaningful values, and the clean typography.

- **Redundant legend:** The legend sits outside the chart area rather than directly labeling the bands. Tufte strongly prefers direct labeling inside the chart, which would eliminate the eye-travel between legend and bands.

4. Graphical Variables Used and Their Suitability

According to Bertin's graphical variables, the visualization employs several visual encoding techniques.

Graphical Variable	Usage	Suitability
Position (X-axis)	Represents years from 1980 to 2023	Highly suitable because time series are best represented spatially along a horizontal axis.
Position (Y-axis)	Represents annual number of deaths	Highly suitable because position is one of the most accurate visual channels for quantitative data.
Color(Value/lightness)	Not used systematically	The warm-to-cool ordering of hues implies a vague age sequence (blue = youngest, orange = oldest), which is intuitive but not explicit
Color Hue	Differentiates age groups	Suitable because distinct colors help identify categories.
Length/Height	Indicates changes in mortality levels over time	Suitable and easy to perceive for the bottom layer.
Ordering	Age groups are stacked consistently across time	Suitable because it supports pattern recognition and trend tracking.

Overall Evaluation of Graphical Variables

The most effective graphical variables used are position and color, which clearly communicate trends and categorical distinctions. The use of area is appropriate for showing contributions to the total number of deaths but is less effective for precise comparisons among age groups. Consequently, while the stacked area chart successfully communicates the overall decline in diarrheal disease mortality, a grouped line chart or multiple small-multiple charts could improve detailed comparisons between age groups.